Deployment Precision Calibration

Frozen HYB-MS +10% ensemble • identity-disjoint calibration only

GATE PASSED

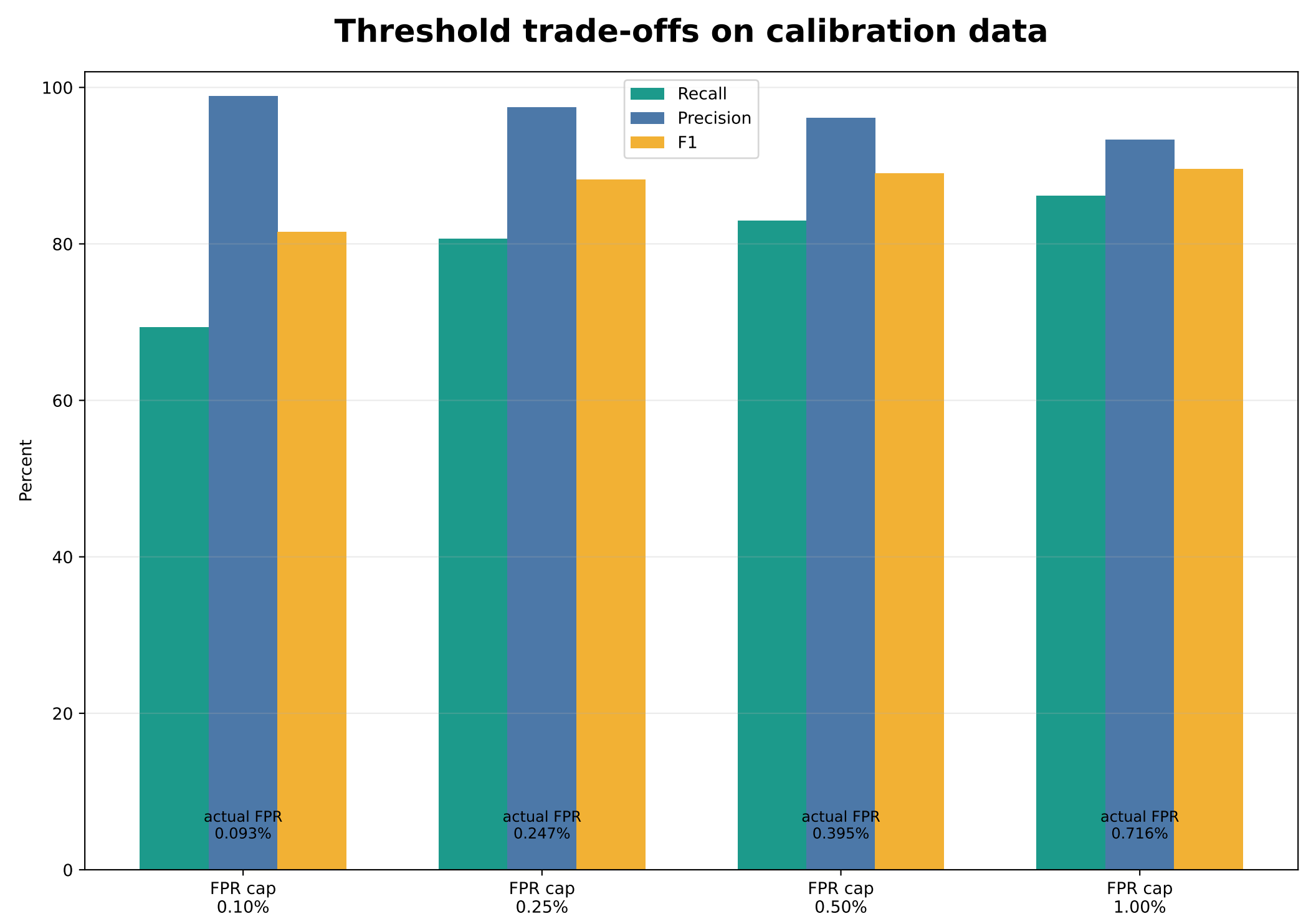
Selected threshold: 0.995117

Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
97.76%	97.44%	80.60%	88.22%	97.54%	93.34%	0.25%	19.40%

Gate: FPR ≤ 0.25% and recall ≥ 80%. Observed: FPR 0.25%, recall 80.60%.

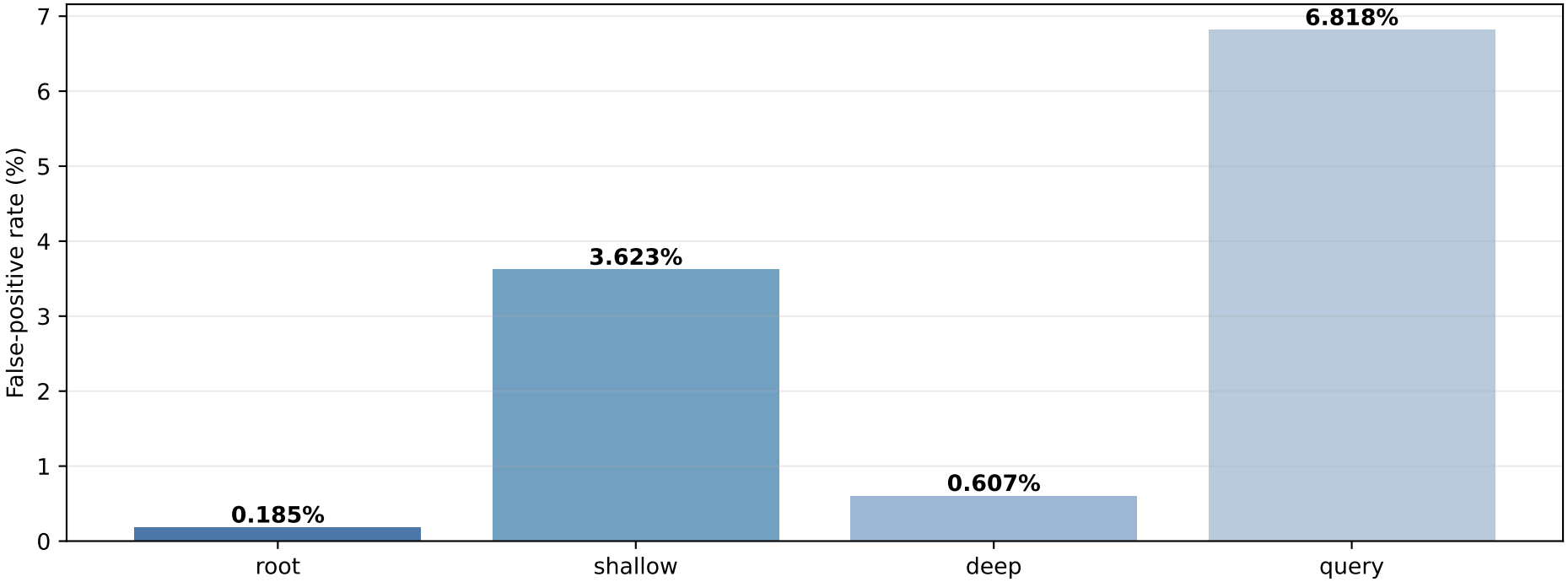
Counts: TN 21,447 • FP 53 • FN 485 • TP 2,015

This threshold was chosen without using the final confirmation set. If the gate fails, the model must be retrained using new development data and recalibrated on a separate cohort.



Original threshold 0.989095: recall 84.48%, FPR 0.58%. Selected deployment threshold is 0.995117.

Where False Positives Occur



Feature-pattern counts among false positives

deep path: 8 • many subdomains: 3 • none of registered flags: 26 • punycode: 14 • security words: 1 • query string: 3

These categories identify likely collection or representation weaknesses; they are diagnostic evidence, not permission to tune against the untouched confirmation set.

Operational Interpretation

At 0.1% phishing prevalence	**projected precision 24.66%**
At 1% phishing prevalence	**projected precision 76.76%**
At 5% phishing prevalence	**projected precision 94.51%**
At 10% phishing prevalence	**projected precision 97.32%**

Retraining decision

Gate passed; preserve the frozen model. Retraining is not justified before untouched confirmation.

Next locked action

Apply this exact ensemble and threshold once to a newly frozen, identity-disjoint low-prevalence confirmation cohort. Do not revise the threshold after viewing those labels.